



Research on X-ray Image Fusion Algorithm for Food Foreign Object Detection

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Abstract

In the field of non-destructive testing of foreign objects in food, the high cost and low efficiency of manual labeling greatly limit the application of X-ray foreign object detection systems. To overcome this problem, this paper proposes a technique for fusing foreign object images with food images, and the fused images enable the automatic labeling of foreign objects. Firstly, X-ray images of foreign objects and food images were collected, and data augmentation was performed on the foreign object images to increase their diversity. Then the food images were fused with the enhanced foreign object images, and the foreign objects were automatically labeled in the fused food images. Finally, the foreign object detection models Model_Y2 and Model_Y1 were established using the dataset automatically annotated by the image fusion method and the dataset manually collected and annotated by traditional methods. The results demonstrate that the proposed method substantially decreases annotation time by 90% while concurrently improving annotation efficiency and accuracy. Comparatively, Model_Y2 outperforms Model_Y1 with a 4.5% higher mAP@0.5:0.95. This indicates that the method not only enhances data annotation efficiency and quality but also improves the accuracy of X-ray foreign object detection, providing a highly efficient and practical technical solution for the intelligent development of food safety inspection.

Keywords Data annotation · Target detection · Image fusion · X-ray · Yolo

1 Introduction

During the food production process, foreign objects such as stones, metals, plastics, and glass may inadvertently be mixed into food. If these foreign objects are ingested by consumers, it will endanger their health and even their lives. In addition, with the development of Internet technology, the

Internet has spread to every corner of the world. Once foreign objects are found in food, it will generate online public opinion and quickly spread, which will have a significant negative impact on food production enterprises. Therefore, food production enterprises have strengthened the inspection of foreign objects in food and spawned many non-destructive testing technologies to detect foreign objects in food.

X-ray technology, a non-destructive testing method, is widely employed in the food industry for detecting foreign bodies due to its superior transmission characteristics [1–7]. X-rays penetrate materials at different rates depending on their thickness and density, producing images with pronounced black and white contrasts. Since foreign bodies typically have distinct densities and shapes compared to the food products themselves, analyzing X-ray images can effectively identify the presence of foreign objects within food [6, 8–11]. The integration of deep learning for detecting foreign bodies in X-ray images represents a significant advancement in the food industry [6, 8–15]. Deep learning excels in image recognition and feature extraction [16, 17]. Training deep learning models for foreign body detection requires a substantial corpus of annotated data, which

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provides the necessary images and information about target locations and categories, enabling the model to accurately identify and localize targets [18–22].

Common methodologies for acquiring annotated data include manual annotation, crowdsourcing, and semi-automatic model annotation [18]. Manual annotation, conducted by individuals or teams within a single organization, is often inefficient and costly [13]. Crowdsourcing, which involves outsourcing tasks to external platforms, boosts efficiency but frequently yields variable annotation quality. This variability stems from differences in the skills, experience, and attention of the annotators, who may interpret annotation guidelines differently, thereby causing inconsistent and inaccurate annotations in certain instances [19]. Semi-automatic model annotation entails using lightweight models to learn from a subset of the data and then automatically annotate the remaining data. Although this approach enhances efficiency, it suffers from significant limitations regarding inference accuracy and adaptability to novel samples. Lightweight models often fail to capture the full complexity and nuances of the data, leading to annotation errors. Additionally, when confronted with new and unseen samples, these models struggle to generalize effectively, resulting in inaccurate annotations that limit their practical utility [19, 20].

Currently, manual annotation remains the predominant approach for obtaining annotated data, but it is time-consuming and labor-intensive, with the efficiency and quality of annotation often falling short of expectations [23–29]. To address this challenge, the study introduces an X-ray-based data augmentation technique for food foreign body images. This technique utilizes generated image annotation data as a substitute for manual annotation, thereby alleviating the burden of manual annotation and reducing the difficulty of acquiring food foreign body annotation datasets [30–34].

2 Materials and Methods

2.1 Test Instrument

In this study, the experimental apparatus employed was the UNX4015-N model food X-ray foreign body detection system, manufactured by UNICOMP. This advanced equipment is specifically engineered for the inspection of small to medium-sized packaged and bagged food products. The technical specifications, which are crucial for understanding the device's operation and effectiveness, are presented in Table 1 [35].

The UNX4015-N ensures superior safety by maintaining an X-ray radiation level below $1\mu\text{Sv/h}$, which facilitates the safe conduct of experiments. Figure 1 provides an actual image of the UNX4015-N. All sample image data used in this experiment were acquired using this device.

Table 1 Technical parameters

Term	Specifications
Size	1200(L) × 960(W) × 1900(H)mm
X-ray source and detector	80kv/210–350 W, 0.2 mm, 0.4 mm, TDI
Operating environment	Relative temperature $-10 \sim 40^{\circ}\text{C}$, relative humidity 30 ~ 90%, no condensation
Electric power source	AC220 V, 1.5 KVA, 50/60 Hz
Conveyor speed	10–60 M/min, 10–40 M/min (heavy-duty)
Radiation safety	$\leq 1\mu\text{Sv/h}$



Fig. 1 UNX4015-N: small and medium-sized packaging/bagged X-ray detection equipment

2.2 Test Data Acquisition and Processing

2.2.1 Background Image Collection

In this investigation, the background images are X-ray images of samples captured by the instrument, which are primarily categorized into two distinct types: NG (No

Good) images and OK (Okay) images. NG images depict samples that have failed inspection due to the presence of extraneous materials, such as stones, metals, plastics, and glass, either within or on the surface of the sample. These foreign objects are not inherent to the sample's composition. In contrast, OK images represent samples that have passed inspection, confirming the absence of any foreign substances. The datasets collected for this experiment are shown in Table 2, with 9 datasets containing a variety of bagged and packaged products and bulk products. A part of the X-ray sample images are presented in Fig. 2.

2.2.2 Foreground Image Acquisition

The foreground image, referred to as a foreign image, is obtained by extracting the area occupied by the foreign object. In the context of food foreign object detection, it is

typically necessary to mark the area of the foreign object in the NG image. The foreign objects, which include stones, metals, plastics, and glass, exhibit certain similarities in shapes, sizes, and specific attributes, allowing them to be classified into six categories, as shown in Table 3.

The collection of Foreign images primarily involves three methodologies: manual annotation, image segmentation, and the utilization of pre-annotated open-source image segmentation datasets.

- Manual annotation requires meticulous and detailed annotation of specific food categories within NG images, facilitating the acquisition of prevalent foreign object images pertinent to those categories [18].
- Conversely, the image segmentation approach leverages sophisticated techniques such as deep learning [16, 17], threshold segmentation [9], and edge detection [11] to

Table 2 Collection of datasets

Datasets	Name	Number	Image size	Product Packaging	Foreign body type
D1	Food Boxes	79	1024*1536	Bulk	Glass, ceramic, steel ball, steel wire
D2	Food Boxes	195	1024*1536	Bulk	Glass, ceramic, steel ball, steel wire
D3	Food Boxes	139	1024*1536	Bulk	Glass, ceramic, steel ball, steel wire
D4	Food Boxes	178	1024*1536	Bulk	Glass, ceramic, steel ball, steel wire
D5	Sausages	594	1024*1024	Bag	Glass, ceramic, steel ball, steel wire
D6	Sausages	401	1024*1024	Bag	Glass, ceramic, steel ball, steel wire
D7	Sausages	146	1024*1024	Bag	Glass, ceramic, steel ball, steel wire
D8	Sausages	300	1024*1024	Bag	Glass, ceramic, steel ball, steel wire
D_Y	Bagged Snacks	400	512*512	Bulk	Glass, ceramic, steel ball, steel wire

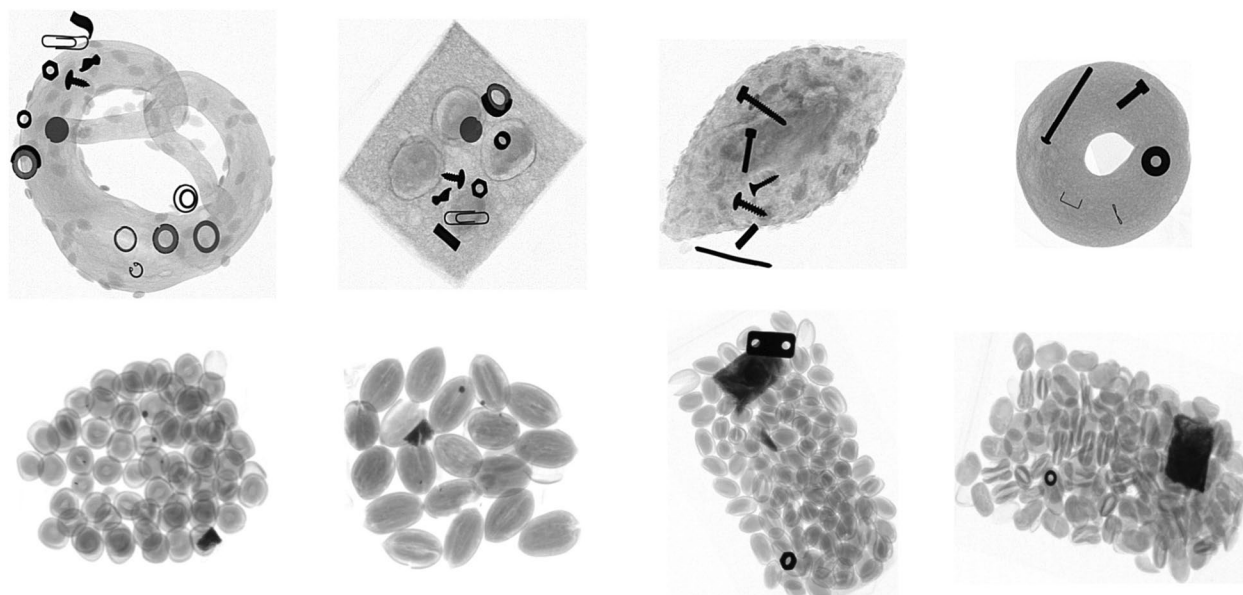


Fig. 2 Partial X-ray sample images: contains conventional metal foreign objects

Table 3 Criteria for classification of foreign bodies

Number	Tag	Type of foreign	Clarification
0	foreign	Punctate	Small dots, short filaments, large dots
1	block	Blocky	The plan view is an irregular polygon
2	longline	Filamentous	Long, thin, curved, and irregular silk threads
3	ring	Cyclic	Concentric circle
4	screw	Screw	Sharp, flat screws, screws
5	shape	Specific	Paper clips, springs, art knives, hair clips

extract Foreign images from existing annotated NG food images.

- Additionally, by employing pre-annotated image segmentation datasets, such as MS COCO, PASCAL VOC 2012, and Cityscapes [17, 18], the segmented Foreign images can be appropriately scaled and repurposed as surrogate sources for foreign object images.

These methodologies are pivotal in mitigating the manual annotation workload while concurrently enhancing the efficiency and quality of foreign object image acquisition [30, 31, 34, 36, 37]. The foreign objects present in the NG images and the acquired Foreign images are depicted in Fig. 3.

2.2.3 Image Fusion Methods

Traditionally, the annotation of foreign objects in NG images relies on manual annotation, which is time-consuming, labor-intensive, and often results in inconsistent quality and efficiency. To address these challenges, this study introduces an innovative X-ray-based data augmentation technique that addresses these issues by effectively combining foreign body images (foreground) with food images (background) to create high-quality synthetic datasets. The detailed image fusion process is depicted in Fig. 4.

Image enhancement techniques are applied to both OK and foreign images before merging the foreground and background images. Multiple Foreign images are integrated with a single OK image, replacing NG food images with synthetic images containing foreign objects. This process involves

several key components: grayscale processing, size adjustment, data augmentation, and placement quantity limitation.

- **Grayscale Processing:** Foreign images undergo grayscale processing based on the specific food type. This involves converting the pixel values of regions outside the contour edges in the Foreign images to 255 if their grayscale values exceed a predefined threshold. This step is essential for minimizing noise impact on the annotation boxes' size in the Foreign images.
- **Image Size Adjustment:** The Foreign images are resized to match the dimensions of the OK images and the actual size of the foreign objects. This adjustment ensures a more natural appearance of the foreign objects within the OK images.
- **Image Data Augmentation:** Both OK and Foreign images are subjected to various operations, such as rotation, cropping, scaling, and transposition, to augment the image data. This augmentation significantly increases the volume of sample data available for analysis.
- **Placement Quantity Limitation:** Parameters are set to control the number of Foreign images placed within a single OK image, with adjustments made according to the type of food. This regulation ensures a balanced integration of foreign objects in the synthetic images.

The automatic annotation process for image data fusion can be divided into two main components: the classification of foreign object markers (C) and the positional data (P) of the target bounding boxes for these objects. The

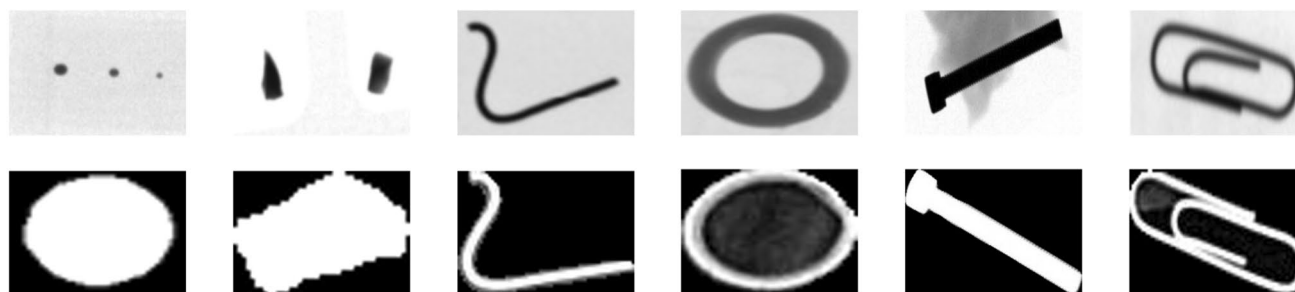


Fig. 3 Foreign object pictures: the foreign object image in the NG image and the extracted foreign object image contour

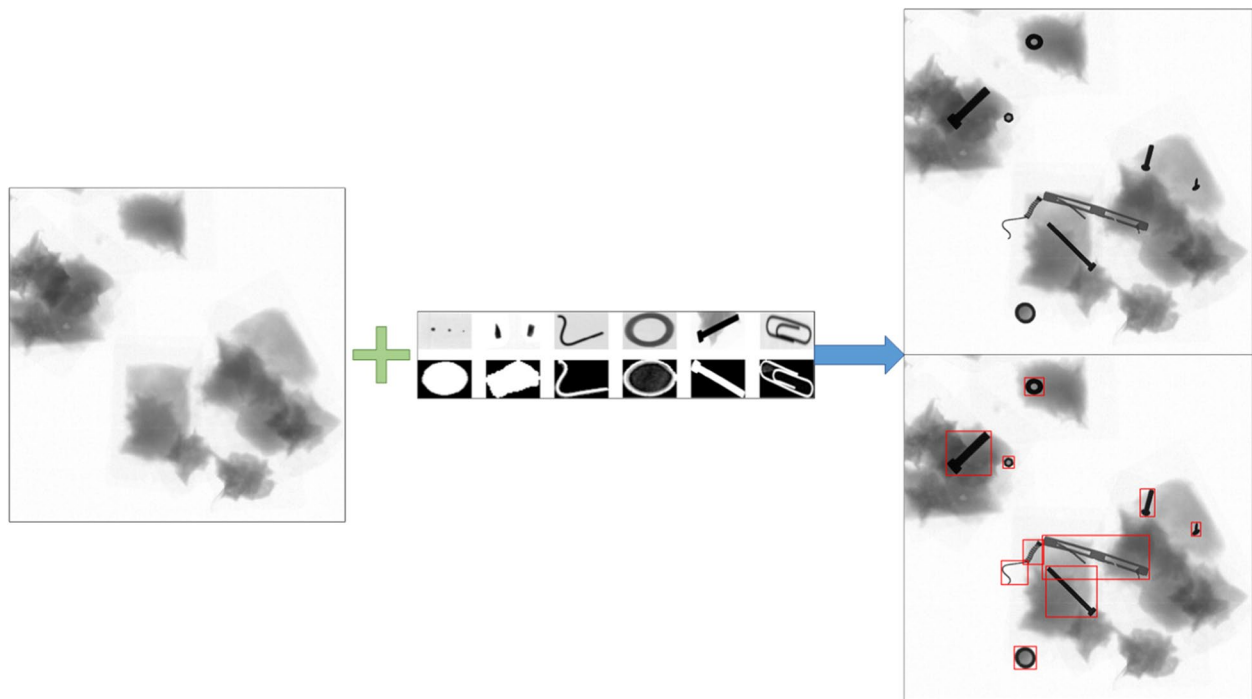


Fig. 4 Image Fusion Process: fusion of foreign object images (foreground) and food images (background) to generate a synthetic dataset

classification component, C , is derived from the naming conventions of foreign object images. Specifically, the substring before the first underscore '_' in the file name is extracted using regular expressions. For example, in an image named 'screen_09-36-54-347_1.png', the label in the annotation file is 'screen', which corresponds to the number 4 in Table 3. The positional data, P , is determined through a series of steps that involve the combination rules of the OK image and the foreign object image.

Step 1, Random Placement, involves specifying the position (x, y) of the upper-left corner of the foreign object image based on the dimensions (w_1, h_1) and (w_2, h_2) of the OK and foreign images, ensuring it remains within the boundaries of the OK image.

$$(x, y) = \begin{cases} x, x \in (0, w_1 - w_2) \\ y, y \in (0, h_1 - h_2) \end{cases} \quad (1)$$

Step 2, Placement Rule, involves calculating the average grayscale value of the placed area on the OK image. If this value exceeds 245, indicating a predominantly white area, the effectiveness of foreign object detection is reduced, necessitating a return to Step 1.

Step 3, Adjustment of Fusion Ratio: The parameter α (0—1, step size 0.1, randomly selected) adjusts the weighted fusion of OK and foreign images. As α

increases, the foreign image's proportion decreases while the OK image's increases, controlling the grayscale of foreign images in composite images [30, 31, 34, 36, 37].

$$\alpha \in (0, 1) \quad (2)$$

$$\text{img} = \text{mask} * (1 - \alpha) + \text{ok} * \alpha \quad (3)$$

Step 4, Inspection Rule, requires calculating the average grayscale value (g_1) of the placed area in the OK image and the minimum grayscale value (g_2) in the foreign image. If $(g_1 < g_2 + g)$ (a debugging value), it suggests that the grayscale of the foreign image is higher than that of the OK image post-processing, making the labeling less meaningful, and necessitating a return to Step 1.

Step 5, Position Calculation, involves calculating the label position information (x_0, y_0, w_0, h_0) based on the random position (x, y) and the size information of the OK and foreign images.

$$(x_0, y_0, w_0, h_0) = \begin{cases} x_0 = (x + w_2/2)/w_1 \\ y_0 = (y + h_2/2)/h_1 \\ w_0 = w_1/w_2 \\ h_0 = h_1/h_2 \end{cases} \quad (4)$$

Step 6, Return Label, in which the label name and position are combined to return a single tuple L representing the foreign object label.

$$L = (C, x_0, y_0, w_0, h_0)$$

(5)

Step 7, Save Labels, a TXT file with the same name as the OK image is created, and the label tuples from Step 6 are appended. If the number of placements N is met, the foreign object placement for this OK image is complete, and the process moves to the next OK image. Otherwise, the process returns to Step 1 to continue placement.

Ultimately, all annotation files are converted from TXT to XML or JSON formats and are then examined with the relevant annotation tools, as illustrated in Fig. 5 [18].

2.3 Test Method

The food foreign object detection model is based on the YOLOv5 framework, which runs on the Windows 10 platform [38, 39]. YOLOv5, with its multi-scale feature network for precise sub-millimeter object detection and lightweight design ensuring real-time production-line inspection, has low hardware requirements for industrial PCs and is widely applied in industrial object detection. Two models were trained based on the images in the D_Y dataset in Table 2: Model_Y1 using D_Y1 (manually labeled packaged snacks dataset) and Model_Y2 using D_Y2 (automatically labeled packaged snacks dataset). The hardware, software, and related parameters of Model_Y1 and Model_Y2 are detailed in Table 4.

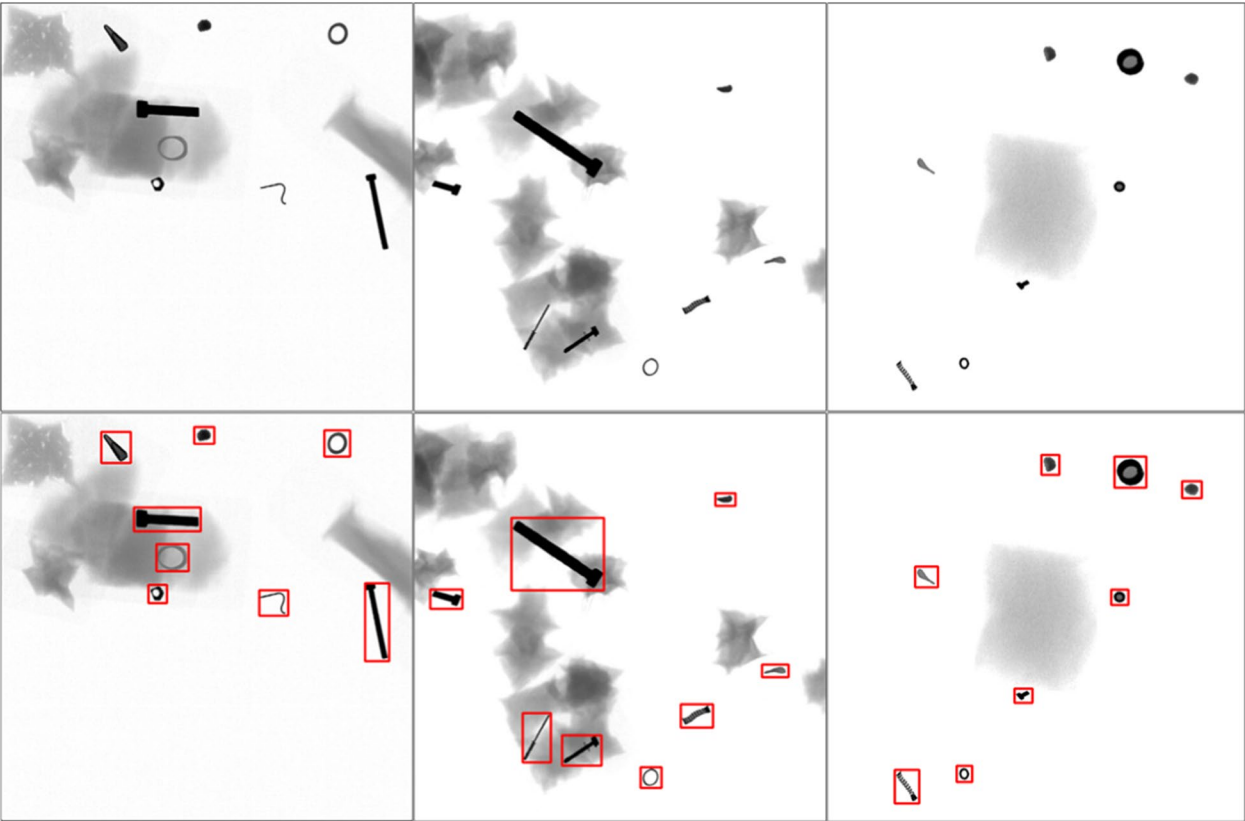


Fig. 5 Annotation effect: original images (up) and an-notated images (down)

Table 4 Model related parameter

Relevant Parameters	Y1(Manual Labeling)	Y2 (This article)
Hardware	Intel(R) Core(TM) i7-10,700 64G + NVIDIA GeForce RTX 3080	
Software	Python 3.12 + Pytorch 2.3.1 + CUDA 12.1 + YOLOv5	
Yolov5	batch-size 8, weights yolov5 s.pt, imgsz 512	
D_Y1	400 images	\
D_Y2	\	400 images

To effectively assess the performance of the food foreign object detection model, we use precision (P), recall (R), average precision (AP), and mean average precision (mAP) to evaluate Model_Y1 and Model_Y2. The formulas for these calculations are as follows.

$$P = \frac{TP}{TP + FP} \quad (6)$$

$$R = \frac{TP}{TP + FN} \quad (7)$$

$$AP = \int_0^1 P(R) dR$$

$$mAP = \frac{\sum_{i=1}^N AP_i}{N} \quad (8)$$

In the formula, TP (True Positives) denotes the number of positive samples that the model correctly predicts. FP (False Positives) represents the count of negative samples that the model incorrectly classifies as positive. Conversely, FN (False Negatives) signifies the number of positive samples that are mistakenly labeled as negative by the model. The variable N refers to the total number of categories within the dataset. AP_i , or average precision for the i -th detection target, measures the model's ability to accurately identify positive samples across various thresholds. Finally, mAP (mean average precision) is the mean of all AP_i values, providing a comprehensive assessment of the model's performance across all categories.

3 Results

3.1 Labeling Efficiency and Quality

In the annotation of X-ray food images, manual annotation typically requires the identification of NG (Not Good) images, which involves a more complex process than annotating OK (Good) images. This complexity arises from the need to manually insert foreign objects into the product, resulting in a time cost that is twice that of acquiring OK images. In contrast, the image enhancement method

proposed in this paper focuses on annotating OK images, offering a more efficient alternative. As detailed in Table 5, the method introduced in this study significantly reduces the time required for image annotation. Specifically, it decreases the workload associated with identifying foreign objects and obtaining images by 50%, and it reduces the workload for image annotation and label verification by 90%. Consequently, this method greatly enhances the efficiency of the image annotation process.

The annotation time is affected by several factors, such as the total number of images, the quantity of targets per image, the dimensions of the images, and the size of the target bounding box, in addition to the varying annotation processes, as illustrated in Table 6. The data augmentation method employed in this article incurs minimal time overhead for automatic annotation. In typical X-ray food foreign object detection datasets, the annotation time is merely 5 min, which is significantly lower than that of manual annotation.

Annotation times across various datasets are illustrated in Fig. 6, as detailed in Table 6. This innovative approach substantially reduces time investment, thereby enhancing annotation efficiency and lowering labor costs.

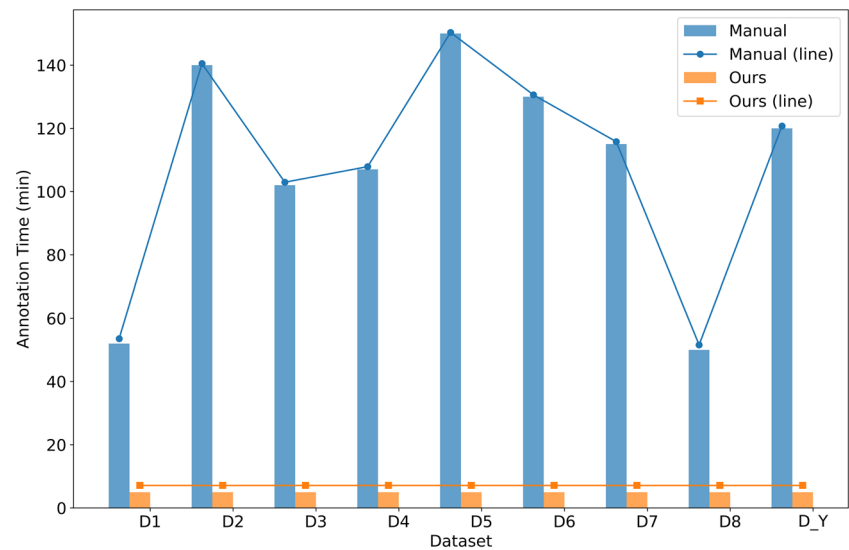
In terms of annotation quality, a comparative analysis of the annotation box positions and sizes is presented in Fig. 7 between the manually annotated dataset (D_Y1) and those generated by our proposed method (D_Y2). The results elucidate that the inherent subjectivity in manual annotation results in D_Y1 producing annotation boxes that are, on average, larger than those generated by D_Y2. This discrepancy poses challenges in maintaining consistent levels of efficiency and accuracy. Furthermore, fluctuations in annotator habits contribute to inconsistencies, rendering it difficult to ensure that each annotation box precisely encapsulates its corresponding foreign object, thus leading to deviations in both position and size. Conversely, D_Y2 utilizes automated annotation techniques, which enable them to accurately discern and delineate the contours of foreign objects, effectively fitting them within the annotation boxes. However, D_Y1 demonstrates enhanced performance in annotating certain minute foreign objects. Notably, in the first and third images of the top row, the number of annotation boxes in D_Y1

Table 5 Labeling process

Labeling process	Manual annotation	This paper	Time consumption ratio
Identify foreign objects	Real foreign object	Select foreign object image	2: 1
Image acquisition	NG image acquisition	OK image acquisition	2: 1
Image preprocessing	Program	Program	1: 1
Image annotation	Manual annotation(D_Y1)	Automatic annotation(D_Y2)	10: 1
Label proofreading	Manual proofreading	program	10: 1

Table 6 Labeling efficiency

Datasets	Number of images and annotation targets/piece			Annotation time/min	
	Number of pictures	Total number of annotations	Number of single icon annotations	Manual annotation	This article
D1(Food Boxes)	79	2688	34	52	5
D2(Food Boxes)	195	8412	43	140	
D3(Food Boxes)	139	5238	37	102	
D4(Food Boxes)	178	10307	57	107	
D5(Sausages)	594	3449	6	150	
D6(Sausages)	401	8956	22	130	
D7(Sausages)	146	3118	21	115	
D8(Sausages)	300	5330	17	50	
D_Y(Bagged Snacks)	400	3073	7	120	

Fig. 6 Labeling time consumption: single person annotation, multi person annotation, and comparison of methods in this article

corresponds accurately to the actual count of foreign objects, whereas D_Y2 occasionally produces multiple annotations for the respective images. D_Y1 employs manual labeling, especially in cases with small foreign objects and complex backgrounds, where human judgment is crucial for precision. As a result, D_Y1 still has certain advantages over automatic labeling methods in detecting small foreign objects.

3.2 Test Results

A comparative analysis of evaluation metrics for two models, Model_Y1 and Model_Y2, on both training and validation sets is provided in Table 7. Model_Y1 was trained using the manually annotated dataset D_Y1, while Model_Y2 was trained with dataset D_Y2, which was automatically annotated using the data augmentation method proposed in this study. Both models were trained for 300 epochs utilizing the model framework. On the training set, Model_Y2 showed a 2.6% lower

recall (R) compared to Model_Y1, which had a 4.6% lower value in this metric. However, Model_Y2 surpasses Model_Y1 in precision (P) and mAP@0.5:0.95 by 7.4% and 4.5%, respectively, with a similar pattern observed on the validation set. These results indicate that although Model_Y2 has a slightly higher miss rate than Model_Y1, it effectively reduces false positives. Overall, Model_Y2 demonstrates superior detection performance on both the training and validation sets compared to Model_Y1, although attention should be given to its missed detection issues.

The line charts of various metrics over the 300 training epochs for Model_Y1 and Model_Y2, respectively, are illustrated in Fig. 8 and Fig. 9. It is evident that Model_Y1 experiences significant fluctuations in the P, R, and mAP@0.5 metrics, and its values for P and mAP@0.5:0.95 were lower than those of Model_Y2. In contrast, Model_Y2 exhibits smaller fluctuations during the training process, indicating better algorithm stability and higher precision in training metrics. In summary, Model_Y2 not only achieves superior detection

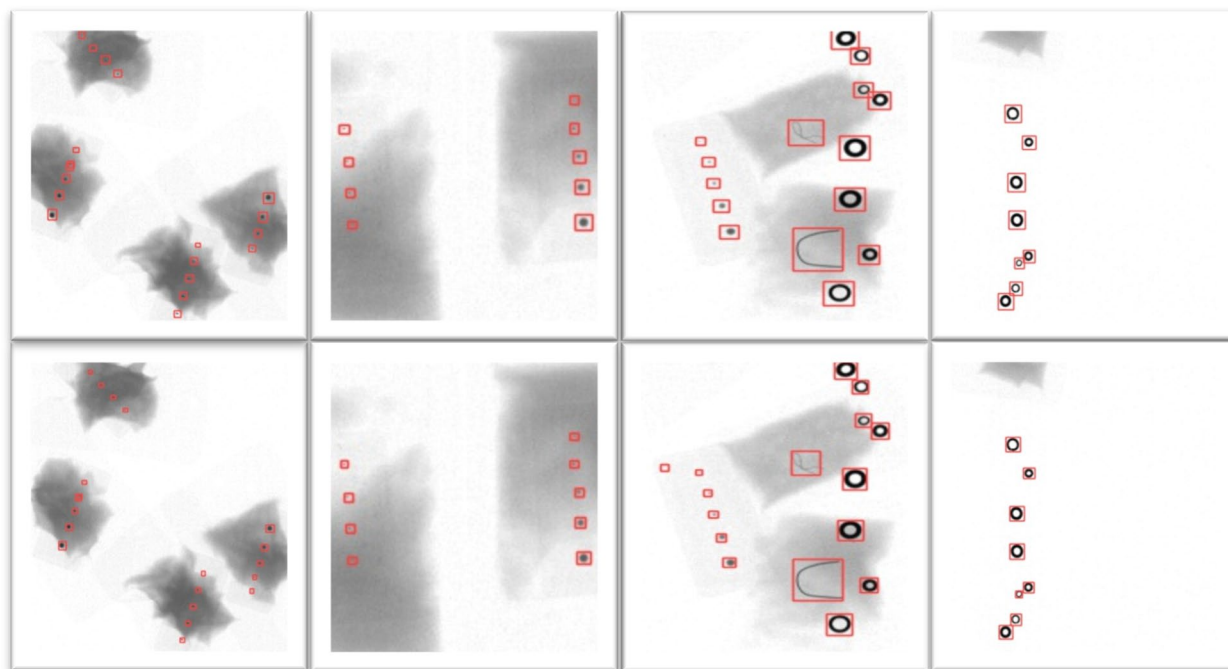


Fig. 7 Image annotation quality: firstly, manually annotate the dataset; secondly, annotate the data augmentation method in this article

4 Discussion

In the domain of X-ray non-destructive food foreign object detection, manual annotation has historically been the predominant method for acquiring image label data. However, this process is both time-consuming and labor-intensive. To address these challenges, our research introduces an innovative data augmentation technique that intelligently combines multiple foreign object images as foregrounds with food OK images as backgrounds. This approach automatically generates a rich dataset of annotated images, thereby significantly reducing the difficulty of constructing X-ray food foreign object datasets. Compared to traditional manual annotation, our method not only streamlines the acquisition and processing workflow of X-ray images but also substantially reduces the time required for annotation, thereby enhancing both the efficiency and quality of the annotation process [26, 40].

Despite these advancements, our automatic annotation technique encounters certain limitations, particularly when dealing with food images that exhibit subtle grayscale differences and when detecting very small foreign objects [7, 33, 41]. To address these challenges, future research will concentrate on two core areas. First, for food images with minimal grayscale differences, we will dynamically adjust the grayscale values of foreign objects based on specific food types and thoroughly consider the various types of foreign objects that may appear during the production process. Second, in the detection of very small foreign objects, we will refine the acquisition strategy for small foreign images. This

Table 7 Model Evaluation Metrics

Dataset	Model	P/%	R/%	maP@0.5/%	maP@0.5:0.95/%
Validation	Model_Y1	90.5	88.2	91.4	46.6
	Model_Y2	98.2	83.5	86.1	51.2
Train	Model_Y1	90.8	87.4	91.5	48.4
	Model_Y2	98.2	84.8	86.9	52.9

performance but also demonstrates enhanced stability and precision in its training metrics compared to Model_Y1.

The detection performance of Models Y1 and Y2, shown in Fig. 10, is analyzed from the fit and accuracy of the detection box. Model_Y2's bounding boxes better match object sizes than Model_Y1's, as the manual annotation of Model_Y1 is subject to individual biases, causing inconsistent box sizing. In terms of detection accuracy, Model_Y2 outperforms Model_Y1, especially for small, dot-shaped foreign objects. As Fig. 10 shows, Model_Y1 misses several detections, with 4 fewer identifications than Model_Y2 in the first row, and a similar shortfall in the second row. Model_Y2's advantage stems from its automated, data-augmented annotation method, which enables it to handle object diversity more effectively and reduce missed detections.

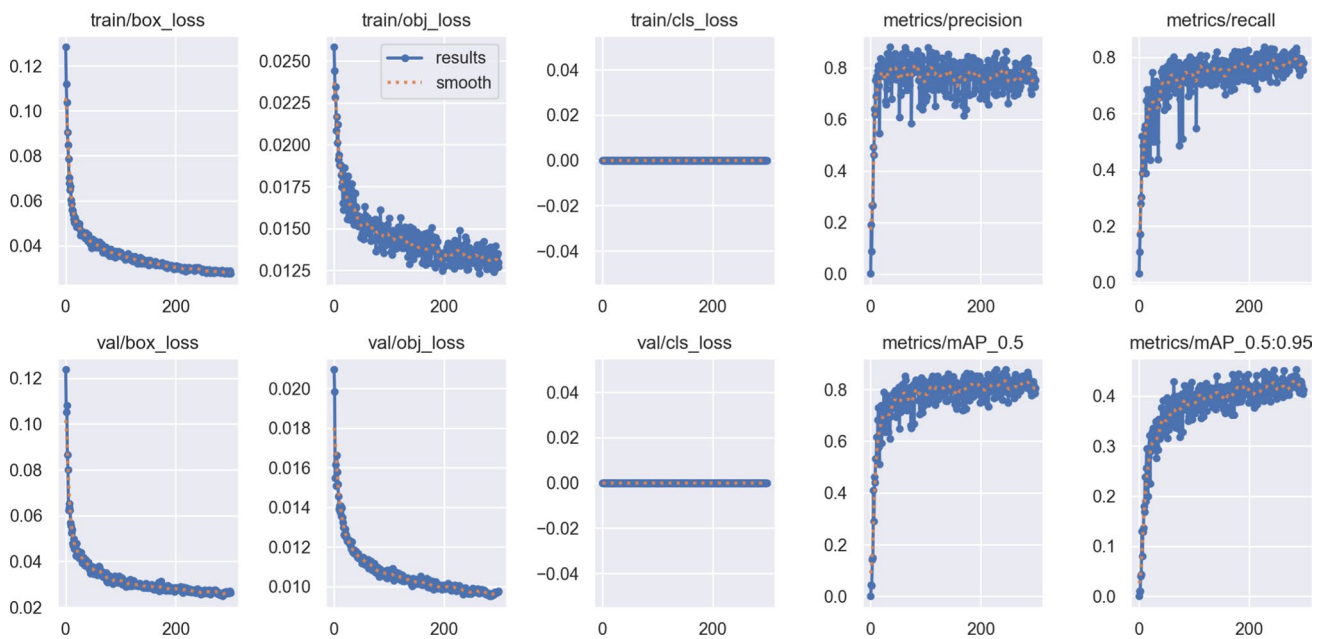


Fig. 8 Model_Y1 training metrics: metrics for each round of model training

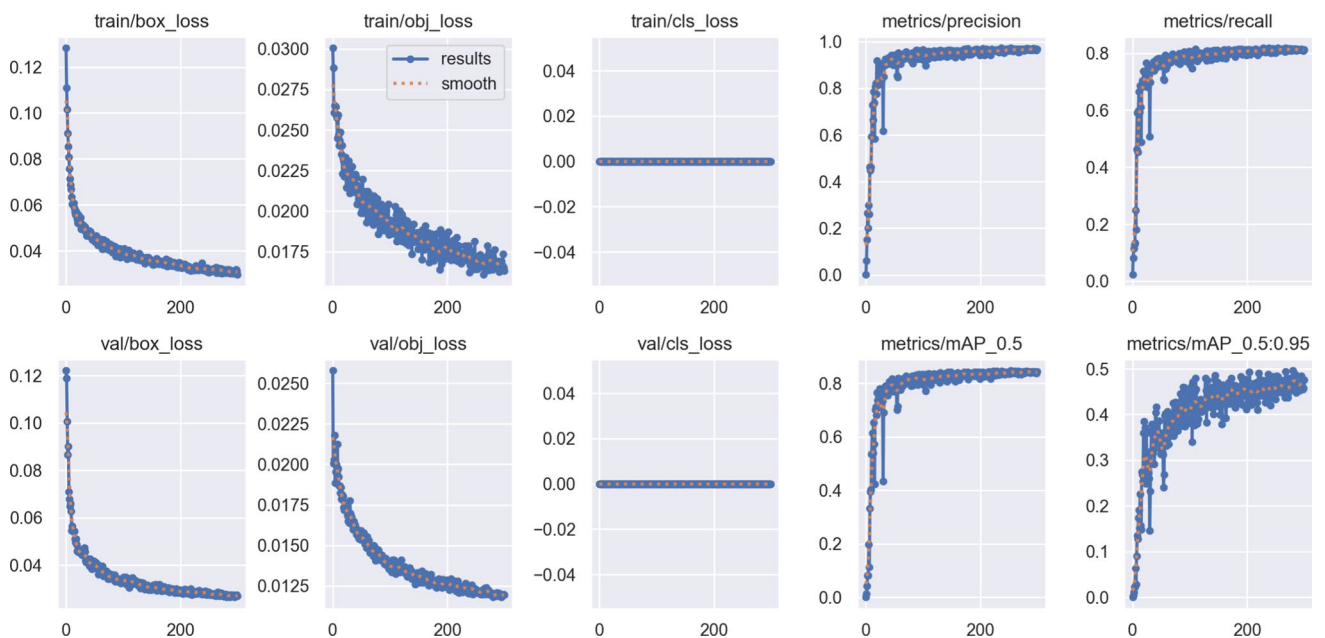


Fig. 9 Model_Y2 training metrics: metrics for each round of model training

refinement will involve not only extracting the contour information of foreign objects but also enhancing the extraction of grayscale information within the contour area, and strategically optimizing the grayscale values of foreign images.

Moreover, to further enhance the authenticity and effectiveness of detection, future research plans to superimpose foreign images on NG (non-qualified) images that already

contain foreign objects [30–34]. This strategy aims to enrich the diversity and complexity of foreign object placement, thereby aligning more closely with actual application scenarios [42–44]. Compared to the current practice of generating synthetic images solely on OK (qualified) images, this approach is anticipated to significantly improve the performance of the detection system.

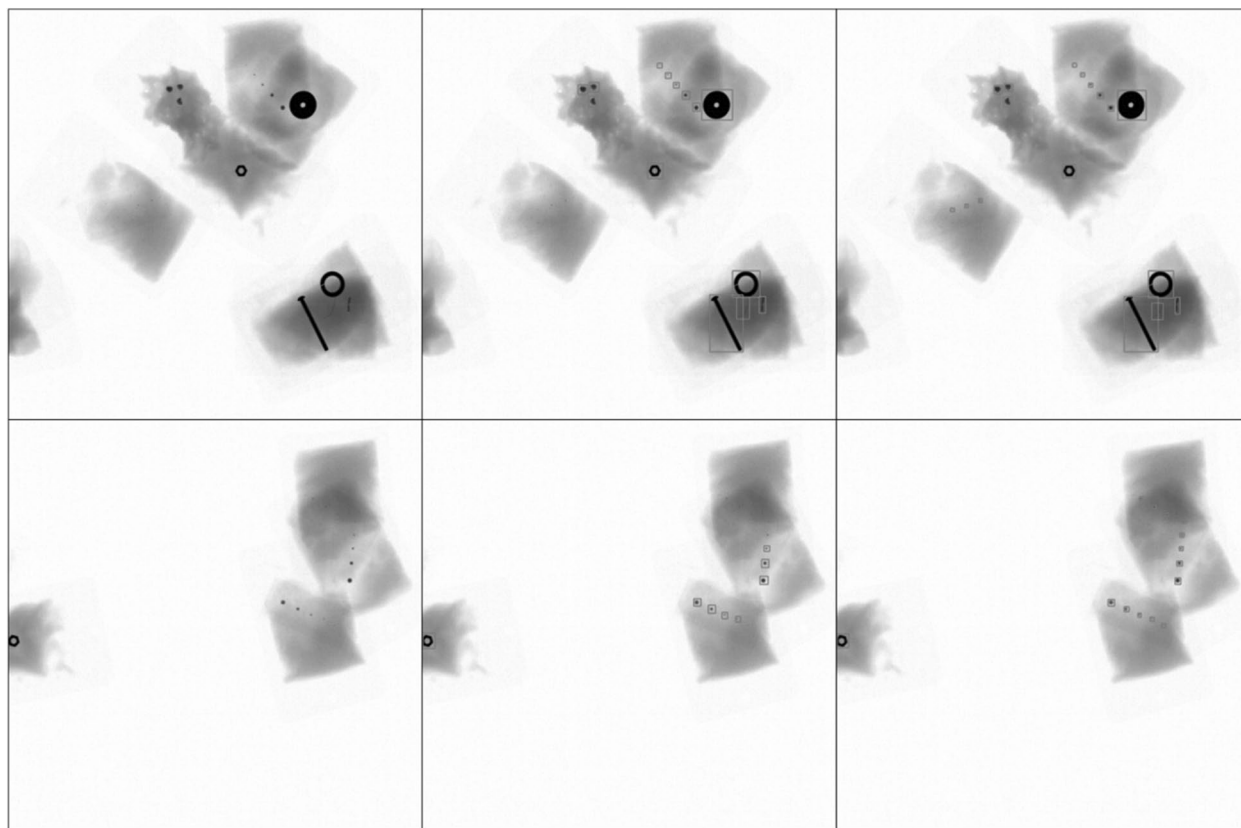


Fig. 10 Visual detection effects of Model Y1 and Model Y2: From left to right, the original image, Model Y1 detection effect, and Model Y2 detection effect are shown in sequence

5 Conclusion

A data enhancement-based X-ray food foreign object detection method is proposed, which generates a food foreign object image annotation dataset by skillfully fusing foreign images (as foreground) with food OK images (as background). In terms of automated image annotation, our method greatly simplifies the process of acquiring and processing X-ray food images, significantly improves annotation efficiency and quality, and effectively reduces reliance on manual annotation. In terms of food foreign object detection performance, compared to the Model_Y1 trained using manually labeled data, the Model_Y2 trained using the dataset generated by our image enhancement method exhibits improvements of 7.4% and 4.5% in P(Precision) and mAP@0.5:0.95 (mean Average Precision) on the training and validation sets, respectively, demonstrating higher detection accuracy. Additionally, the Model_Y2 displays less fluctuation in indicators during the training process, indicating better stability. In summary, the method proposed in this study not only achieves significant results in reducing the labeling cost of food foreign matter datasets but also substantially improves labeling efficiency and quality. This has

great practical significance and application value, promoting the development of the food foreign matter detection industry.

Author Contributions Jianfeng Yao and Zhenyang Wu wrote the main manuscript text. Pengtao Wang and Junchao Ye revised manuscript. Jingxian Wang prepared Figs. 1, 2, 3, 4 and 5. All authors reviewed the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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